

Single image haze removal using dark channel prior

N8EN11D Projet Image / Image Processing project



Figure 1: The dehazing algorithm proposed by [2] removes the haze in the original image (left), generating a haze-free image (center) and a depth map of the scene (right) from the estimated transmission map.

1 Introduction

Images captured in outdoor environments are often degraded by atmospheric phenomena such as haze, fog, or smoke. These effects reduce visibility, lower contrast, and alter colors, making subsequent image processing or computer vision tasks more difficult.

The paper “*Single Image Haze Removal Using Dark Channel Prior*” [2] proposes a method to restore a clear image from a **single hazy input image**, without requiring multiple views or additional information.

The method relies on a simple but interesting observation called the **dark channel prior**: in most natural (haze-free) images, within a local neighborhood, at least one color channel tends to have very low intensity values. In hazy images, this property is weakened due to the presence of atmospheric light.

Using this prior, the method estimates two contribution terms in the image formation, the **transmission map** (*i.e.* how much light is attenuated due to haze), and the **atmospheric light**, which accounts for the global light of the scene.

These estimates are then used within a physical image formation model to recover the haze-free image.

2 Project Objectives

2.1 Implementation

The first step will be to read the paper to understand the theoretical foundations of the Dark Channel Prior.

- Implement the algorithm in a programming language of your choice (C++, Python, or Matlab is preferable).
- Ensure the code is **well-structured and modular**, with proper **decomposition into functions**. Your implementation must follow good software engineering practices:
 - Modular design (separate functions for each step of the pipeline)
 - Clear decomposition of the algorithm:
 - * Dark channel computation
 - * Atmospheric light estimation
 - * Transmission estimation
 - * Transmission refinement

- * Image reconstruction
- * Depth map reconstruction (up to a scale factor)
- Proper documentation and comments
- Use the **same parameters as described in the paper** (e.g. patch size, ω , minimum transmission threshold, etc.) and avoid hardcoding inputs.
- During testing, your implementation should be easy to run with **any given image and a set of user-defined parameters**. Your program should accept any input image and parameter configuration (e.g., via command line), and possibly be run in batch mode over a set of images.
- *While not required, optimizing the implementation with **parallelization techniques** (e.g., *OpenMP* for C++, *parfor* for MATLAB, etc.) could be a **bonus**— but only if you are familiar with these techniques, as the main focus should remain on the implementation and evaluation.*

2.2 Experimental Evaluation

This is the **most important part of the project**. You must conduct a thorough evaluation of your implementation using both **qualitative** analysis (visual results) and **quantitative** metrics such as SSIM [5], PSNR (c.f. [4, eq. 2.119]), or other relevant metrics.

You should investigate:

- Influence of parameters (patch size, ω , etc.)
- Behavior on different types of scenes (urban, natural, sky regions)
- Failure cases (e.g., bright objects, sky)

Here are some public datasets providing hazy images with their relevant ground truth images:

- RESIDE Dataset [3], includes synthetic hazy images with ground truth;
- O-HAZE Dataset [1], real haze with corresponding haze-free images;

Note that the method [2] is targeting outdoor scenes, but it would still be interesting to see its limitations in dealing with indoor scenes.

References

- [1] Codruta O. Ancuti et al. *O-HAZE: a dehazing benchmark with real hazy and haze-free outdoor images*. URL: <https://data.vision.ee.ethz.ch/cvl/ntire18/o-haze/>.
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- [3] Boyi Li et al. *RESIDE: A Benchmark for Single Image Dehazing*. URL: <https://data.vision.ee.ethz.ch/cvl/ntire18/o-haze/>.
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- [5] Zhou Wang et al. “Image quality assessment: from error visibility to structural similarity”. In: *IEEE Transactions on Image Processing* 13.4 (Apr. 2004), pp. 600–612. ISSN: 1941-0042. DOI: 10.1109/tip.2003.819861. URL: <http://dx.doi.org/10.1109/TIP.2003.819861>.